



ReLoop
CIRCULAR WASTE

Give every piece of waste a better destination.

ReLoop identifies waste with AI, connects households to verified collectors, and keeps a measured record of what was actually recovered.

Team TechGaint

SANKALP Climate Edition

01

Scan an item

02

Understand it

03

Get it collected

04

Prove it was recovered

Most waste is thrown away on a guess.

Recycling fails long before the facility. It fails at the moment of decision — when nobody can answer a simple question about the thing in your hand.

01

What is this material, really?

02

Can it be recycled at all — or is it hazardous?

03

Where do I take it, and does it accept my material?

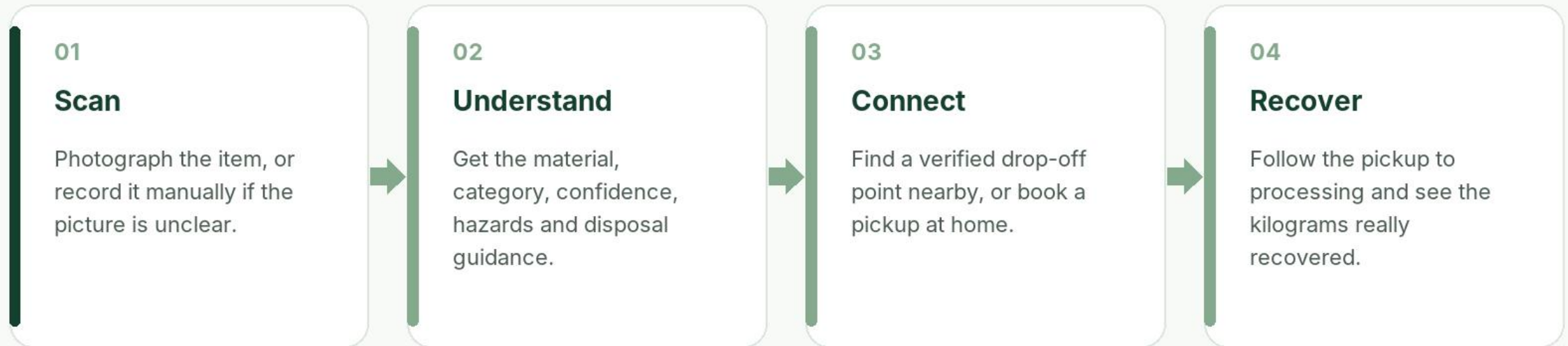
04

What actually happened after it was collected?

A household cannot act on an answer it never gets — so material that could loop back goes to landfill instead.

Photograph it, then follow it to recovery.

Four steps, one loop, and a real weight at the end of it.



Every stage is enforced server-side: a pickup cannot jump from requested to recovered, and a collector can only act on work assigned to their own organisation.

Three surfaces, one loop.

All three are built and working today — this is a product walkthrough, not a mock-up.

AI waste scan

The backend calls Gemini and returns a structured reading: item, category, confidence, recyclability, hazards and a disposal instruction. If the provider is unavailable the app says so and offers manual classification — it never invents a result.

Collection points

A directory and map of verified drop-off points with the materials each one accepts, opening hours and distance from the user. Verified and unverified records are labelled differently and never merged.

Pickups & tracking

Book a pickup, then follow a server-enforced lifecycle from requested through accepted, scheduled, weighed, processing and recovered. The resident sees the weight a collector recorded on site.

One account type for households, one for collectors, one for administrators — with authorisation enforced on every record, not just on the route.

Only measured weight counts.

Most platforms report what was collected. ReLoop separates what was measured from what was estimated — and never mixes the two.

Measured

Kilograms a collector weighed on site, collection counts, per-material history and every completed collection. These are facts about the account and are never smoothed, extrapolated or estimated.

Estimated

Avoided CO₂e, derived from conservative published per-material coefficients. Always labelled as an estimate wherever it appears, and documented in full so a judge can check every coefficient.

If nothing has been collected yet, the impact page says so instead of showing a number. There are no seeded statistics anywhere in the product.

Built for real households and real collectors.

The parts of a waste platform that decide whether people trust it.

Vetted collectors

A collector can act only after an administrator verifies their application. Verification is what grants the collector role at all.

Private by default

The open pool shows material, city and a kilometre-rounded distance. The address, exact coordinates, notes and photo unlock only after the request is accepted.

Safe uploads

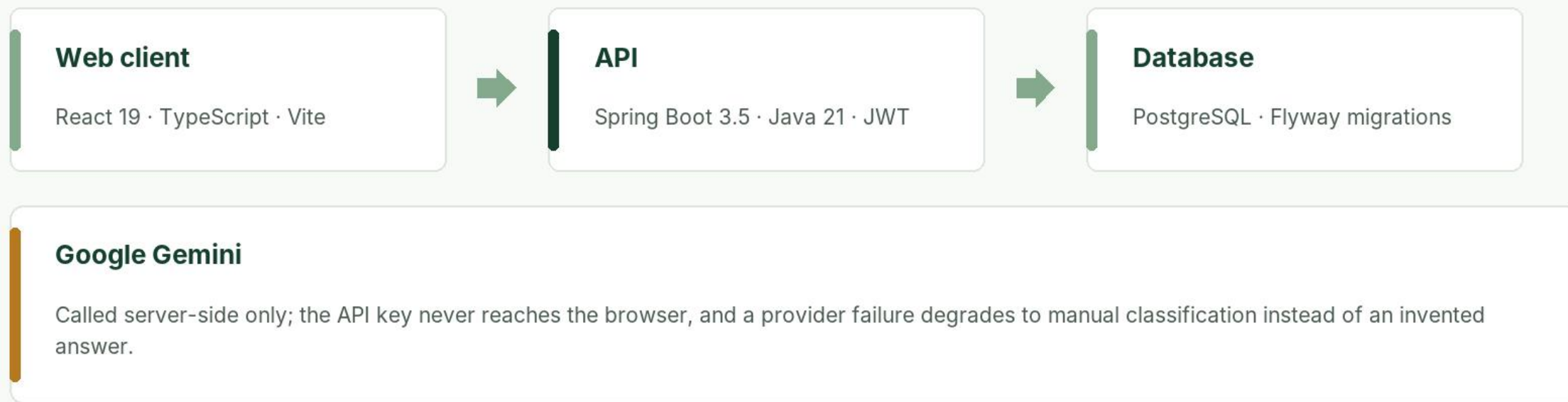
Images are validated by content — JPEG, PNG or WEBP, size-capped — and unsupported or corrupted files are refused before any provider call.

Race-safe claims

Two collectors cannot both win the same job: the second gets a clean 409 and is told to refresh, never a phantom assignment.

A production-shaped stack, not a prototype.

Three tiers, one outward dependency, and a test suite that runs against the real thing.



90 automated tests run against a real PostgreSQL schema and the real HTTP stack — no mocked success paths.

One-command deployment with Docker Compose, plus CI that runs the full suite and the client build.

Numbers a judge can check.

Every figure is either a measured weight or a labelled estimate with a published method.

What we report

Measured kilograms and completed collections, plus per-material breakdowns and a dated recycling history. Avoided CO₂e is shown separately and always labelled an estimate.

The methodology document lists every coefficient, its rationale and its limitations — transport emissions are not subtracted, and hazardous waste is credited at zero because no certified route is verified yet.

kg CO₂e avoided per kg

conservative, literature-based

Plastic	1.5
Paper	1.0
Metal	4.0
Glass	0.3
Hazardous	0.0

From one city to a verified network.

The build already separates the things that must change at scale.

One city pilot

Single instance, local image storage, the full pickup lifecycle and admin console. Deployable today with a single command.

Now

Many collectors, one pool

Geospatial indexing for the pool, a shared rate-limit store for multiple replicas, object storage behind the existing interface, and partner onboarding at scale.

Next

A verified recovery network

Deeper proof of downstream recovery, region-specific emission factors, and an open partner API for facilities to report outcomes back.

Then

Demo it in three minutes.

Everything listed is implemented and covered by the test suite — nothing is staged.

- ✓ Authentication with JWT and rotating refresh tokens
- ✓ Collection-point directory with distance search
- ✓ Collector application and admin verification
- ✓ Recycling history and measured impact
- ✓ AI waste scan, with an honest manual fallback
- ✓ Full pickup lifecycle, enforced server-side
- ✓ Notifications for every state change
- ✓ Admin console: users, partners, catalog, analytics

The demo path

docker compose up → scan an item → book a pickup → a verified collector weighs it → mark it recycled → read the measured impact and history.

Pilot ReLoop with your network.

We have the platform. A pilot needs one city and a route for the material.

What we need

One city, a handful of verified collectors, and a partner willing to receive the material we recover. We handle onboarding, vetting and reporting.

What you get

A working platform, honest measured reporting with a documented method, and a repeatable one-command deployment you can host yourself.

Code, tests, migrations, Docker setup and documentation are all in the repository — this is an engineering result, not a concept.

Give every piece of waste a better destination.

A measured record of waste returning to circulation — not a promise that it was recycled.

Team TechGaint

Thank you · github.com/Pavan3030-pr/Reloop